

TY B.Sc. IT – IKS in IT

Question Bank with Answers

This PDF contains the questions and the answers from the supplied IKS in IT question bank. The original MCQ options are retained.

Unit 1

1. The term Pramana refers to: A) A mathematical formula■ B) A means of obtaining valid knowledge■ C) A programming language■ D) A binary number

Answer: B) A means of obtaining valid knowledge

2. Which Pramana means knowledge gained through direct perception? A) Shabda■ B) Anumana■ C) Pratyaksha■ D) Upamana

Answer: C) Pratyaksha

3. Which Pramana is based on logical reasoning? A) Upamana■ B) Shabda■ C) Pratyaksha■ D) Anumana

Answer: D) Anumana

4. Knowledge obtained through comparison is called: A) Shabda■ B) Upamana■ C) Anumana■ D) Pratyaksha

Answer: B) Upamana

5. Which Pramana refers to knowledge obtained from reliable verbal testimony? A) Shabda■ B) Upamana■ C) Pratyaksha■ D) Anumana

Answer: A) Shabda

6. Which school of philosophy is mainly associated with logical reasoning? A) Yoga■ B) Nyaya■ C) Sankhya■ D) Buddhism

Answer: B) Nyaya

7. Which school primarily emphasizes knowledge of Brahman and ultimate reality? A) Vedanta■ B) Nyaya■ C) Mimamsa■ D) Charvaka

Answer: A) Vedanta

8. The Mimamsa school mainly focuses on: A) Astronomy■ B) Vedic rituals and interpretation■ C) Medicine■ D) Architecture

Answer: B) Vedic rituals and interpretation

9. Which feature of IKS is closely related to modern information systems? A) Random reasoning■ B) Structured knowledge representation■ C) Guesswork■ D) Storytelling

Answer: B) Structured knowledge representation

10. Rule-based frameworks in Indian traditions are most similar to: A) Computer games■ B) Rule-based Expert Systems■ C) Graphic designing■ D) Computer hardware

Answer: B) Rule-based Expert Systems

11. Traditional Indian knowledge contributes to modern AI mainly through: A) Painting techniques■ B) Knowledge representation and reasoning■ C) Music composition■ D) Animation

Answer: B) Knowledge representation and reasoning

12. Algorithmic thinking refers to: A) Solving problems randomly■ B) Following systematic step-by-step procedures■ C) Memorizing formulas■ D) Drawing diagrams

Answer: B) Following systematic step-by-step procedures

13. The decimal number system developed in ancient India is based on: A) Base-2■ B) Base-8■ C) Base-10■ D) Base-16

Answer: C) Base-10

14. The place value system is important because: A) Every digit has a fixed value■ B) A digit's value depends on its position■ C) Numbers have no order■ D) It uses Roman numerals

Answer: B) A digit's value depends on its position

15. Which Indian mathematician is credited with defining arithmetic rules involving zero? A) Panini■ B) Aryabhata■ C) Pingala■ D) Brahmagupta

Answer: D) Brahmagupta

16. In modern computing, binary numbers are represented using: A) 0 and 2■ B) 1 and 2■ C) 0 and 1■ D) 1 and 10

Answer: C) 0 and 1

17. The concept of Shunya (Zero) is important in computing because: A) It replaces decimal numbers■ B) It forms the basis of binary logic■ C) It increases memory size■ D) It is used only in mathematics

Answer: B) It forms the basis of binary logic

18. Pingala's Chandas Shastra is mainly associated with: A) Astronomy■ B) Sanskrit grammar■ C) Binary representation of poetic meters■ D) Medicine

Answer: C) Binary representation of poetic meters

19. Pingala's work is considered an early example of: A) Computer networking■ B) Combinatorics and binary enumeration■ C) Database management■ D) Cloud computing

Answer: B) Combinatorics and binary enumeration

Unit 2

20. Which modern field is inspired by the systematic reasoning found in Indian Knowledge Systems? A) Artificial Intelligence■ B) Graphic Design■ C) Video Editing■ D) Web Browsing

Answer: A) Artificial Intelligence

1. Aryabhata is best known for his contributions to: A) Medicine■ B) Grammar■ C) Mathematics and Astronomy■ D) Architecture

Answer: C) Mathematics and Astronomy

2. Aryabhata's computational methods mainly involve: A) Random calculations■ B) Step-by-step computational procedures■ C) Trial-and-error methods only■ D) Graphical representation

Answer: B) Step-by-step computational procedures

3. Aryabhata's methods for calculating square roots and cube roots are examples of: A) Data compression ■ B) Iterative computational algorithms ■ C) Binary coding ■ D) Database searching

Answer: B) Iterative computational algorithms

4. Which of the following best describes algorithmic thinking? A) Memorizing formulas B) Solving problems through systematic, logical steps C) Guessing the answer D) Using only computers

Answer: B) Solving problems through systematic, logical steps

5. Which programming concept is most similar to Aryabhata's repeated computational procedures? A) Loop (Iteration) B) Variable Declaration C) Comments D) Array

Answer: A) Loop (Iteration)

6. The Nyaya school is mainly concerned with: A) Music B) Logical reasoning and valid knowledge C) Medicine D) Architecture

Answer: B) Logical reasoning and valid knowledge

7. Which of the following is NOT one of the five parts of the Indian syllogism? A) Proposition B) Reason C) Example D) Assumption

Answer: D) Assumption

8. In Nyaya syllogism, Hetu refers to: A) Conclusion B) Proposition C) Reason D) Example

Answer: C) Reason

9. Which part of the Indian syllogism provides a commonly accepted illustration to support the reasoning? A) Pratijna B) Udaharana C) Nigamana D) Upanaya

Answer: B) Udaharana

10. The final conclusion in the Nyaya syllogism is called: A) Hetu B) Upanaya C) Nigamana D) Udaharana

Answer: C) Nigamana

11. Nyaya logic differs from Aristotelian logic because Nyaya: A) Uses binary numbers B) Uses a five-part syllogism with contextual reasoning C) Does not use logical reasoning D) Uses only mathematical calculations

Answer: B) Uses a five-part syllogism with contextual reasoning

12. Aristotelian logic mainly consists of: A) Two parts B) Three-part syllogism C) Four parts D) Five parts

Answer: B) Three-part syllogism

13. Which type of inference predicts an effect from a known cause? A) Sheshavat B) Purvavat C) Samanyatodrishta D) Pratyaksha

Answer: B) Purvavat

14. Which type of inference determines the cause from an observed effect? A) Purvavat B) Sheshavat C) Samanyatodrishta D) Shabda

Answer: B) Sheshavat

15. Which type of inference is based on general observation and repeated experience? A) Purvavat B) Sheshavat C) Samanyatodrishta D) Upamana

Answer: C) Samanyatodrishta

16. Who authored the Arthashastra? A) Aryabhata B) Panini C) Kautilya (Chanakya) D) Brahmagupta

Answer: C) Kautilya (Chanakya)

17. Which of the following is NOT a principle of strategic decision-making described in the Arthashastra? A) Collect information B) Analyze alternatives C) Ignore risks D) Monitor outcomes

Answer: C) Ignore risks

18. Which modern IT activity most closely follows Kautilya's decision-making process? A) Software project planning B) Playing computer games C) Typing documents D) Installing software

Answer: A) Software project planning

19. In an expert system, the Inference Engine is responsible for: A) Storing data permanently B) Applying logical rules to derive conclusions C) Displaying graphics D) Managing network connections

Answer: B) Applying logical rules to derive conclusions

20. Which modern computing field is most closely related to Nyaya logic? A) Computer Graphics B) Artificial Intelligence and Expert Systems C) Web Designing D) Multimedia

Answer: B) Artificial Intelligence and Expert Systems

Unit 3

1. Who is the author of the Ashtadhyayi? A) Aryabhata B) Panini C) Kautilya D) Brahmagupta

Answer: B) Panini

2. Panini's Ashtadhyayi consists of approximately: A) 500 rules B) 1,000 rules C) 2,500 rules D) Nearly 4,000 rules

Answer: D) Nearly 4,000 rules

3. Panini's grammar is considered a rule-based linguistic system because it: A) Uses random word formation B) Generates language using systematic rules C) Has no grammatical structure D) Focuses only on pronunciation

Answer: B) Generates language using systematic rules

4. A meta-rule in Panini's grammar is: A) A rule for pronunciation only B) A rule that controls the application of other rules C) A mathematical formula D) A rule for sentence translation

Answer: B) A rule that controls the application of other rules

5. Rule hierarchy in Panini's grammar helps to: A) Increase vocabulary B) Decide which rule has priority when multiple rules apply C) Translate languages D) Store words in memory

Answer: B) Decide which rule has priority when multiple rules apply

6. Phonetics is the study of: A) Sentence meaning B) Speech sounds C) Programming syntax D) Data structures

Answer: B) Speech sounds

7. Morphology deals with: A) Computer networks B) Word formation and structure C) Database design D) Binary numbers

Answer: B) Word formation and structure

8. Panini's grammar is closely related to which computer science concept? A) Computer Graphics B) Formal Language Theory C) Operating Systems D) Cloud Computing

Answer: B) Formal Language Theory

9. In Formal Language Theory, production rules are used to: A) Store files B) Generate valid strings or sentences C) Compress data D) Encrypt information

Answer: B) Generate valid strings or sentences

10. Which concept is common to both Panini's grammar and modern programming languages? A) Random execution B) Rule-based generation C) Hardware design D) Networking

Answer: B) Rule-based generation

11. Generative grammar mainly focuses on: A) Data transmission B) Deriving words and sentences using rules C) Image processing D) Database management

Answer: B) Deriving words and sentences using rules

12. Which feature of Sanskrit makes it suitable for computational applications? A) Ambiguity B) Precision and unambiguity C) Lack of grammar D) Flexible syntax

Answer: B) Precision and unambiguity

13. Which programming concept resembles Panini's rule-based grammar? A) Sequential execution of instructions B) Random memory allocation C) Hardware assembly D) File compression

Answer: A) Sequential execution of instructions

14. Which phase is the first phase of a compiler? A) Syntax Analysis B) Semantic Analysis C) Lexical Analysis D) Code Generation

Answer: C) Lexical Analysis

15. Lexical Analysis converts a program into: A) Parse Tree B) Machine Code C) Tokens D) Flowcharts

Answer: C) Tokens

16. Syntax Analysis (Parsing) checks whether: A) Variables are initialized B) A program follows grammatical rules C) Memory is available D) The program is executed correctly

Answer: B) A program follows grammatical rules

17. A Parse Tree is created during: A) Lexical Analysis B) Syntax Analysis C) Code Generation D) Linking

Answer: B) Syntax Analysis

18. Natural Language Processing (NLP) enables computers to: A) Manufacture hardware B) Understand and process human languages C) Design computer chips D) Build databases

Answer: B) Understand and process human languages

19. Which of the following is an application of NLP? A) Machine Translation B) Disk Formatting C) CPU Scheduling D) Network Routing

Answer: A) Machine Translation

20. Sanskrit grammar is useful in AI because of its: A) Random sentence structure B) Rule-based and systematic grammatical framework C) Absence of syntax D) Dependence on regional dialects

Answer: B) Rule-based and systematic grammatical framework